

# Hybrid-NL2SVA

**Part 1:** Improving the Hybrid-NL2SVA pipeline

**Part 2:** An end-to-end Spec2SVA framework

Hybrid-NL2SVA Project

New York University

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Hybrid-NL2SVA: Integrating RAG and Finetuning for LLM-based NL2SVA, MLCAD 2025.

# Outline

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Motivation

Pipeline Architecture

Evaluation Setup

Results: RAG+SOR Pipeline

Motivation

AssertionForge: Spec2NL

Integration Architecture

UART Pilot

Results

Next Steps

Part I

# The Hybrid-NL2SVA Pipeline

# Why NL2SVA is hard

Hardware verification can take up to 70% of chip design time

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SVAs are small checkers built into the design: they watch the hardware run and flag it the moment real behavior breaks a rule from the spec. Writing them by hand has two steps: (1) find the rules to check, in plain English, and (2) turn each rule into an SVA – step (2) is **NL2SVA**, and it is what this project is about.

## Why a general AI model struggles.

- Not much SVA code in the data these models learned from
- SVA has timing words that look almost the same but mean different things:
  - $a \mid \rightarrow b$ : b same cycle as a.  $a \mid \Rightarrow b$ : b *1 cycle later*
  - $a \#\#2\ b$ : b *exactly* 2 cycles later.  $a \#\#[1:3]\ b$ : b *anywhere* in cycles 1–3
- Input descriptions can be written at very different abstraction levels of detail, for the same rule:
  - *high-level idea*: “the handshake should work correctly”
  - *some detail*: “that ack correctly follows a request”
  - *fully precise*: “whenever req rises, ack must be asserted exactly 1 clock cycle later”

the first two need grounding first (Step 1, next slide)

# Our approach: three steps

Never trust one AI answer

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**Step 1:** rewrite the input into **OL-NL** (operator-level natural language) – name only real signals (e.g. req, ack, clk, rstn), and describe how they relate in plain words that match one specific operator, e.g. “*whenever req rises, ack must be asserted exactly 1 clock cycle later*” matches `$rose(req) |-> ##1 ack`, operator for operator, in plain English

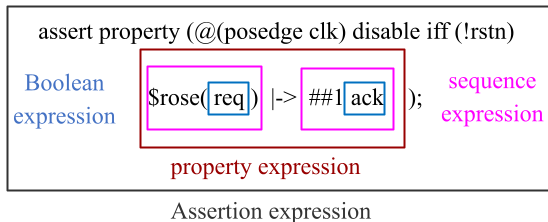
**Step 2:** HybridRetrieval-Augmented Generation – give the model the right operator knowledge, then write a first answer

**Step 3:** use JasperGold to fix syntax errors; also parse the generated SVA into a syntax tree to automatically catch and fix mistakes in what it actually checks

# The structure of an SVA

Four nested layers, each with its own operators – the source of most confusion

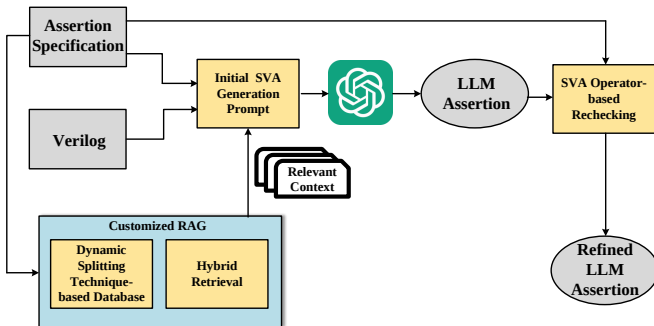
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A Boolean expression (`req`) is wrapped by a sequence expression (`$rose(req) |-> ##1 ack`), wrapped by a property expression, wrapped by the assertion statement itself – getting any one layer's operator wrong (e.g. a sequence operator used where a property operator belongs) silently changes the property's meaning rather than raising a syntax error.

# The original pipeline (MLCAD'25 paper)

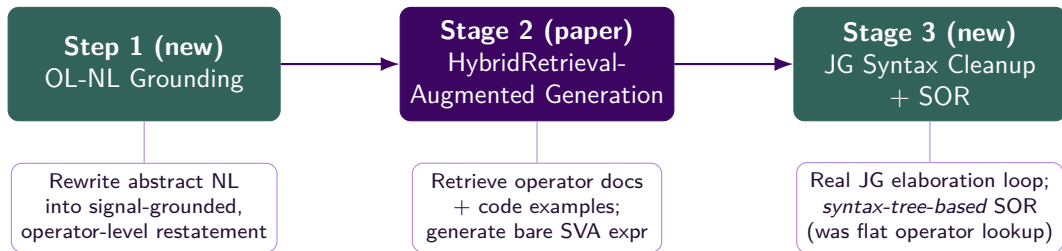
The customized RAG framework, from the MLCAD'25 paper



*Customized RAG* bundles two techniques: **dynamic splitting** (builds a code-centric retrieval database from SVA textbooks, preserving each code snippet with its surrounding explanation) and **Hybrid Retrieval** (next slide). Retrieved context enriches the Initial SVA Generation Prompt; the output is then passed through SVA Operator-based Rechecking (Stage 3) for a refined final assertion.

# The pipeline today: three stages

Step 1 and Stage 3's SOR are both new – only Stage 2 is unchanged from the paper



Step 1 is only needed when the input isn't already signal-grounded. Stage 3's SOR now parses the candidate SVA into a syntax tree and rebuilds its meaning bottom-up, replacing the paper's flat per-operator lookup.

# Stage 1 in detail: what is OL-NL?

Only needed when the input doesn't already name real signals

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**OL-NL = operator-level natural language:** a plain-English sentence with two rules – (1) name *only* real signals from the design, never a made-up name, and (2) say exactly how those signals relate (e.g. “at the same time” or “N cycles later”); each such relation matches one specific SVA operator, but stays plain words, not code.

**Worked example** (same signals as the earlier SVA):

Plain description: that ack correctly follows a request.

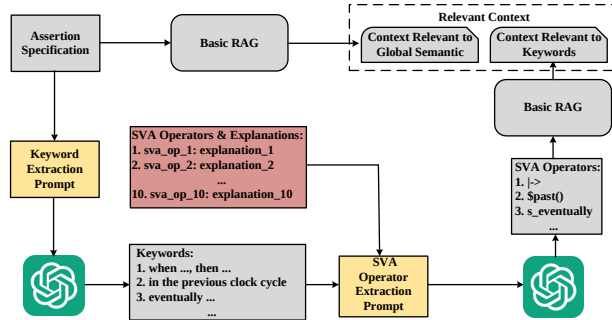
Real signals: req, ack, clk, rstn

OL-NL: Whenever req rises, then exactly 1 clock cycle later, ack must be asserted.

Names only real signals, states the exact timing – matches `$rose(req) |-> ##1 ack` operator for operator, in plain English.

# Stage 2 in detail: HybridRetrieval-augmented generation

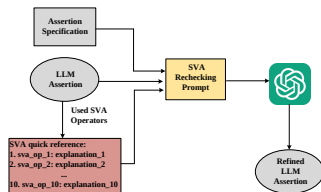
Two independent retrieval paths feed one generation call – from the MLCAD'25 paper



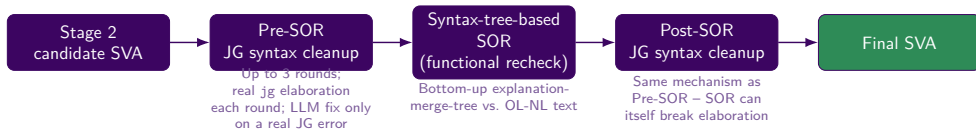
**Path 1 (top):** Basic RAG on the full assertion specification – global-semantic similarity against the code database. **Path 2 (bottom):** an LLM decomposes the description into keyword phrases (*Keyword Extraction Prompt*), maps each to the closest of 10 SVA operators (*SVA Operator Extraction Prompt*), then a second Basic RAG pass retrieves chunks specifically discussing those operators. Both paths' **Relevant Context** feeds the single generation call.

## Stage 3 / SOR: then vs. now

As published (MLCAD'25 paper): flat, per-operator lookup



Today: full Pre-SOR / SOR / Post-SOR pipeline, syntax-tree-based



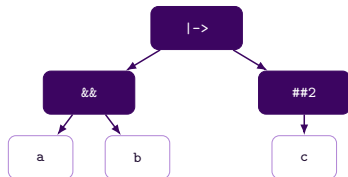
# Inside SOR: the explanation-merge-tree

Parse to a syntax tree, merge bottom-up into NL, then compare against the original ask

LLM-generated SVA (Stage 2):  $a \ \&\& \ b \ \rightarrow \ \#\#2 \ c$

Original ask – OL-NL (Stage 1): whenever a and b are both asserted, then c is asserted 3 cycles later

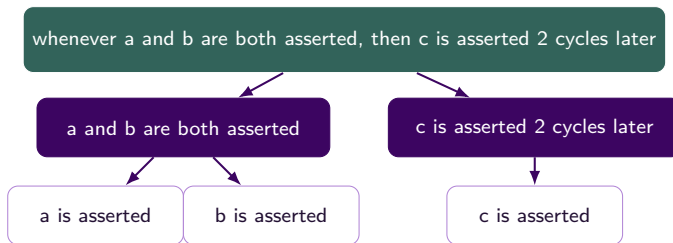
## 1. Parse into a syntax tree



root `|->` splits into `a && b`

and `##2 c`; each splits again

## 2. Merge that same tree bottom-up into NL



→ 3. Compare with the original ask above – mismatch found: the tree's root says 2 cycles later, but the original ask says 3 – SOR flags just the `##2` node and revises it to `##3`, not the whole property.

# Benchmark & metrics

**FVEval-Verified** (wyt2000/FVEval-Verified): a human-expert-corrected version of FVEval's nl2sva\_human and nl2sva\_machine benchmarks.

- **nl2sva\_human\_verified** (73 rows) – expert-written ground-truth SVAs and NL descriptions  
e.g. “that if the current cycle is neither a jump operation nor immediately after a reset pulse, the counter must not overflow” →  
`(!jump_vld_d1 && !tb_reset_1_cycle_pulse_shadow) |-> (count <= max)`
- **nl2sva\_machine\_verified** (283 rows) – rule-generated SVAs, LLM-written NL descriptions; bare dummy testbenches, no reset signal  
e.g. “if both sig\_G and sig\_C are high and sig\_A is high, then sig\_I must not be high 5 clock cycles later” → `(sig_G && (sig_C && sig_A)) |-> ##5 !sig_I`

**Comparison work:** QiMeng-CodeV-SVA: Training Specialized LLMs for Hardware Assertion Generation via RTL-Grounded Bidirectional Data Synthesis, DAC 2026 – a model finetuned specifically for NL2SVA; shown alongside general models as a raw-output baseline.

**Metrics** (via a real JasperGold prop\_eq\_checker): **SC** – does the candidate elaborate against the real testbench on its own? **FM-strict** – formally proven full equivalence to the golden SVA.

# Results (nl2sva\_human\_verified)

Our pipeline vs. raw model output vs. CodeV-SVA

| Model                                 | Configuration  | SC     | FM-strict    | $\Delta$ |
|---------------------------------------|----------------|--------|--------------|----------|
| gpt-4o                                | Raw (0-shot)   | 94.5%  | 63.0%        |          |
|                                       | + Our pipeline | 98.6%  | <b>74.0%</b> | +11.0    |
| DeepSeek-V4-Flash (thinking)          | Raw (0-shot)   | 98.6%  | 76.7%        |          |
|                                       | + Our pipeline | 100.0% | <b>86.3%</b> | +9.6     |
| deepseek-chat (thinking off)          | Raw (0-shot)   | 94.5%  | 54.8%        |          |
|                                       | + Our pipeline | 100.0% | <b>74.0%</b> | +19.2    |
| qwen3-8b (Aliyun)                     | Raw (base)     | 89.0%  | 35.6%        |          |
|                                       | + Our pipeline | 91.8%  | 35.6%        | +0.0     |
| qwen3-14b (Aliyun)                    | Raw (base)     | 93.2%  | 49.3%        |          |
|                                       | + Our pipeline | 91.8%  | <b>65.8%</b> | +16.5    |
| gpt-5 (+ our pipeline, no raw run)    |                | 100.0% | <b>82.2%</b> | –        |
| CodeV-SVA-8B (DAC 2026, specialized)  |                | 97.3%  | 75.3%        | –        |
| CodeV-SVA-14B (DAC 2026, specialized) |                | 93.2%  | 74.0%        | –        |

Best per model family in **bold** (out of 73 rows). DeepSeek-V4-Flash + our pipeline (86.3%) already beats both CodeV-SVA sizes (75.3% / 74.0%), a model finetuned specifically for NL2SVA.  
Hybrid-NL2SVA

# Results (nl2sva\_machine\_verified)

## Our pipeline vs. raw model output vs. CodeV-SVA

| Model                                  | Configuration                         | SC               | FM-strict             | $\Delta$ |
|----------------------------------------|---------------------------------------|------------------|-----------------------|----------|
| gpt-4o (+ our pipeline, no 0-shot run) | –                                     | 99.6%            | 84.8%                 | –        |
| DeepSeek-V4-Flash (thinking)           | Raw (0-shot)<br>+ <b>Our pipeline</b> | 100.0%<br>100.0% | 90.8%<br><b>92.9%</b> | +2.1     |
| deepseek-chat (thinking off)           | Raw (0-shot)<br>+ <b>Our pipeline</b> | 97.5%<br>98.9%   | 78.4%<br><b>85.5%</b> | +7.1     |
| qwen3-8b (Aliyun)                      | Raw (base)<br>+ <b>Our pipeline</b>   | 87.6%<br>88.7%   | 51.2%<br><b>60.8%</b> | +9.6     |
| qwen3-14b (Aliyun)                     | Raw (base)<br>+ <b>Our pipeline</b>   | 91.9%<br>95.1%   | 65.7%<br><b>81.6%</b> | +15.9    |
| CodeV-SVA-8B (DAC 2026, specialized)   | –                                     | 96.5%            | 83.0%                 | –        |
| CodeV-SVA-14B (DAC 2026, specialized)  | –                                     | 97.2%            | 80.9%                 | –        |

Best per model family in **bold** (out of 283 rows). DeepSeek-V4-Flash + our pipeline (92.9%) is the strongest result overall, ahead of both CodeV-SVA sizes (83.0% / 80.9%). gpt-5's machine run was interrupted (45/283) and is omitted.

Part II

# End-to-End Spec2SVA

# From ready-made benchmarks to real designs

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## What Part I evaluated:

NL2SVA on FVEval-Verified – properties and RTL are already picked out and ready to use, one property per row, always paired with a golden (known-correct) SVA.

## What real verification looks like:

a *specification document* (PDF, dozens of pages) and a *multi-file RTL design* – no pre-extracted property list, no golden SVA to check against.

**Goal:** close the loop – *Spec2NL* (turn a spec + RTL into a list of natural-language verification properties) feeding directly into *our* NL2SVA pipeline, with JasperGold verifying the result end to end.

# AssertionForge (NVLabs, LAD 2025)

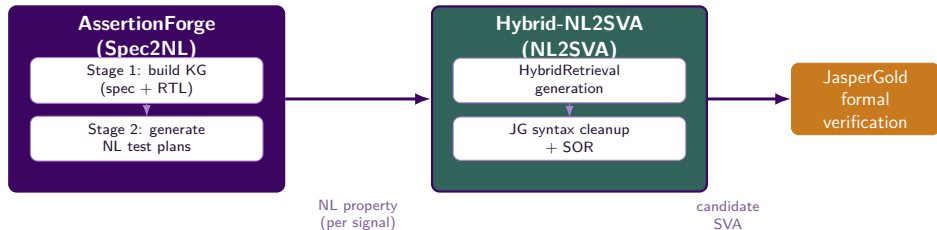
[github.com/NVLabs/AssertionForge](https://github.com/NVLabs/AssertionForge) — [arXiv:2503.19174](https://arxiv.org/abs/2503.19174)

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- **Stage 1 – Knowledge Graph Construction:** builds a joint KG from the spec (via GraphRAG, domain-customized entity-extraction prompt) and the RTL (pyverilog-based structural analysis), then fuses the two
- **Stage 2 – Test Plan Generation:** for each architectural signal, retrieves KG-guided context (RAG chunks + graph-path analysis), prunes with an LLM judge, and generates natural-language verification *plans*

Stage 1 + Stage 2 together = **Spec2NL**: spec PDF + RTL directory → a list of natural-language properties, each grounded in a specific signal and its role in the design.

# Spec2NL + NL2SVA = end-to-end Spec2SVA



NL2SVA runs with **the same settings as the `nl2sva_machine_verified` evaluation**, minus Step 1 (AssertionForge's plans are already signal-grounded) – plus support for a real design's actual clock name (rarely just `clk`).

# Case study: UART

AssertLLM's spec + RTL

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## Design.

- 6 Verilog modules: `uart2bus_top` (real top), `uart_top`, `baud_gen`, `uart_rx`, `uart_tx`, `uart_parser`
- 246 total signals; 18 architectural (interface) signals across the hierarchy
- Real clock port name: `clock` (not `clk`)

## Spec2NL output.

- Knowledge graph: 1,371 nodes, 1,620 edges (spec + RTL fused)
- **323** natural-language verification properties generated (all 18 signals)
- QiMeng-CodeV-SVA's own Table 5 reports **265** SVAs for UART (GPT-4o Spec2NL + GPT-4o NL2SVA) – same order of magnitude

**A real gap found & fixed along the way:** upstream's own signal list example only listed 2 of the 18 real signals, and its LLM-client code shipped as an empty "add your own client" stub – both needed fixing before Stage 1/2 ran end to end.

# Sample generated NL properties

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- **baud\_clk:** “that after a reset condition, baud\_clk will resume toggling without skipping cycles based on the previously set value of baud\_freq”
- **baud\_freq:** “that baud\_freq maintains a constant value when the global\_clock\_freq and baud\_rate inputs remain unchanged over 20 clock cycles”
- **int\_req / int\_gnt:** bus request/grant handshake properties derived from path analysis between uart2bus\_top's internal bus interface signals

Example NL2SVA output (clock\_signal=clock, no disable iff, matching the machine-dataset config):

```
asrt: assert property (@(posedge clock)
    (new_tx_data && !tx_busy) |-> tx_data
);
```

# Validated: same 44 properties, ungrounded vs. grounded

`baud_clk/baud_freq` pilot – the exact same starting ideas, matched one to one

| Same 44 properties, different generation               | n         | #SynC             | #Proven           |
|--------------------------------------------------------|-----------|-------------------|-------------------|
| Raw ideas + gpt-4o 0-shot baseline (no pipeline)       | 44        | 30 (68.2%)        | 15 (34.1%)        |
| Raw ideas + CodeV-SVA-8B (no pipeline)                 | 44        | 18 (40.9%)        | 12 (27.3%)        |
| Raw ideas + CodeV-SVA-14B (no pipeline)                | 44        | 21 (47.7%)        | 13 (29.5%)        |
| <b>Grounded (Step 2 + filter) + full Hybrid-NL2SVA</b> | <b>44</b> | <b>40 (90.9%)</b> | <b>24 (54.5%)</b> |

Same 44 properties both ways (matched one to one) – grounding + the full pipeline is **+22.7pp #SynC**, **+20.4pp #Proven** over the best ungrounded baseline; a specialized NL2SVA model alone (CodeV-SVA) can't make up for bad, made-up signal names the way grounding does. *Scope: `baud_clk/baud_freq` only (2/18 signals) – a small test before the remaining 16 signals and a full re-run.*

## Next steps

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- Run the two-step generation + mechanical filter on the remaining 16 signals and fully regenerate `nl_plans_uart.txt`
- Extend the pilot to the remaining 4 designs from QiMeng-CodeV-SVA's Table 5 (APB, ETHMAC, OPENMSP430, SOCKIT)
- Swap AssertionForge's own NL2SVA backend for a head-to-head comparison at the *same* Spec2NL properties
- Try DeepSeek-V4-Flash / qwen3 as AssertionForge's own Spec2NL backend, not just gpt-4o
- Look into why #SynC (69.7%) trails #SVA – categorize the elaboration failures the same way Part I's row-by-row diffing did

**Thank you**